

Image-Identifiable Genomic Subspaces: A Linear–Gaussian Model of Radiogenomic Recoverability

Radiogenomic Information-Theoretic Bounds Project

May 16, 2026

Abstract

We develop a linear–Gaussian generative model linking patient-level genomic profiles $g \in \mathbb{R}^p$ to imaging phenotypes $x \in \mathbb{R}^d$, and use it to ask a sharper question than “does imaging predict gene j ?” Instead we ask *which low-dimensional genomic subspaces are image-identifiable*. Under the model every quantity of interest is available in closed form: the Bayes-optimal estimator $\mathbb{E}[g \mid x]$, the posterior covariance $\Sigma_{G|X}$, the mutual information $I(g; x)$, and a per-direction *recoverability score* $R(v) \in [0, 1]$. We show that the directions of maximal recoverability solve a generalized eigenproblem that coincides exactly with canonical correlation analysis (CCA) between the two modalities, and that the recoverability of the i -th canonical genomic direction equals the squared canonical correlation ρ_i^2 . We then address estimation in the regime $p \gg n$ via regularized covariances, reduced-rank regression, and the probabilistic-CCA latent-variable form, and describe cross-validated and permutation-based procedures for honest inference. The companion notebook `notebooks/radiogenomic_recoverability.ipynb` implements the full pipeline on a pair of `AnnData` objects, with a synthetic-data fallback that has known ground truth.

1 Setup and notation

We observe n patients. For patient i let $g_i \in \mathbb{R}^p$ be a vector of genomic features (e.g. expression, somatic mutation burden, copy-number scores) and $x_i \in \mathbb{R}^d$ a vector of imaging phenotypes (radiomic descriptors, deep features, or structured reads). Stack the observations into data matrices $\mathbf{G} \in \mathbb{R}^{n \times p}$ and $\mathbf{X} \in \mathbb{R}^{n \times d}$, with row i equal to $g_i^\top$ and $x_i^\top$ respectively. Random variables are written in uppercase (G, X), realizations in lowercase (g, x). Throughout we assume both modalities have been centered, so all means are zero; recentering is handled by the estimators of Section 10.

We use $\Sigma_G \in \mathbb{R}^{p \times p}$, $\Sigma_X \in \mathbb{R}^{d \times d}$ for the marginal covariances, $\Sigma_{GX} = \text{Cov}(G, X) \in \mathbb{R}^{p \times d}$ for the cross-covariance, and $\Sigma_{XG} = \Sigma_{GX}^\top$. For a symmetric positive semidefinite S we write $S^{1/2}$ for its symmetric square root and $S^{-1/2}$ for the inverse thereof.

2 The linear–Gaussian generative model

The genomic state is drawn from a Gaussian prior and the imaging phenotype is a noisy linear readout of it:

$$G \sim \mathcal{N}(0, \Sigma_G), \tag{1}$$

$$X \mid G \sim \mathcal{N}(AG, \sigma^2 I_d), \quad A \in \mathbb{R}^{d \times p}. \tag{2}$$

The matrix A encodes how genomic variation is expressed in the image; the isotropic term $\sigma^2 I_d$ collects everything in the image that genomics does not drive — biological variation outside the measured panel, acquisition and reconstruction noise, and feature-extraction error. Equivalently,

$$X = AG + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d), \quad \varepsilon \perp G. \quad (3)$$

Equations (1)–(3) are deliberately the simplest model with a non-trivial answer. Their value is that *every* downstream quantity is closed-form, so the model functions as an exact upper bound: no estimator — linear or nonlinear — can recover genomics from imaging better than the posterior derived below. Section 10 relaxes the isotropic-noise and known-parameter assumptions for practical fitting.

3 Marginal and joint laws

Proposition 1 (Joint Gaussian law). *Under (1)–(3),*

$$\begin{pmatrix} G \\ X \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A \Sigma_G & A \Sigma_G A^\top + \sigma^2 I_d \end{pmatrix}\right). \quad (4)$$

In particular,

$$X \sim \mathcal{N}(0, \Sigma_X), \quad \Sigma_X = A \Sigma_G A^\top + \sigma^2 I_d, \quad (5)$$

and the cross-covariance is

$$\Sigma_{GX} = \text{Cov}(G, X) = \Sigma_G A^\top. \quad (6)$$

Proof. Since

$$X = AG + \varepsilon,$$

where

$$G \sim \mathcal{N}(0, \Sigma_G), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d),$$

and G and ε are independent, the stacked vector

$$\begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} G \\ AG + \varepsilon \end{pmatrix}$$

is jointly Gaussian, as it is an affine transformation of independent Gaussian variables.

It therefore suffices to compute its mean and covariance. Since both G and ε are centered,

$$\mathbb{E}[G] = 0, \quad \mathbb{E}[\varepsilon] = 0,$$

we obtain

$$\mathbb{E}[X] = A\mathbb{E}[G] + \mathbb{E}[\varepsilon] = 0,$$

and hence

$$\mathbb{E} \begin{pmatrix} G \\ X \end{pmatrix} = 0.$$

Next, the covariance matrix has block form

$$\text{Cov} \begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} \text{Cov}(G, G) & \text{Cov}(G, X) \\ \text{Cov}(X, G) & \text{Cov}(X, X) \end{pmatrix}.$$

The upper-left block is simply

$$\text{Cov}(G, G) = \Sigma_G.$$

For the upper-right block, using $X = AG + \varepsilon$,

$$\text{Cov}(G, X) = \text{Cov}(G, AG + \varepsilon).$$

By linearity of covariance,

$$\text{Cov}(G, X) = \text{Cov}(G, AG) + \text{Cov}(G, \varepsilon).$$

Since G and ε are independent,

$$\text{Cov}(G, \varepsilon) = 0,$$

and therefore

$$\text{Cov}(G, X) = \text{Cov}(G, AG).$$

Using the identity

$$\text{Cov}(U, BV) = \text{Cov}(U, V)B^\top,$$

we obtain

$$\text{Cov}(G, AG) = \Sigma_G A^\top.$$

Hence

$$\Sigma_{GX} = \text{Cov}(G, X) = \Sigma_G A^\top.$$

The lower-left block is the transpose:

$$\text{Cov}(X, G) = \text{Cov}(G, X)^\top = A\Sigma_G.$$

Finally, for the lower-right block,

$$\text{Cov}(X, X) = \text{Cov}(AG + \varepsilon).$$

Expanding the covariance,

$$\text{Cov}(X, X) = \text{Cov}(AG) + \text{Cov}(\varepsilon),$$

since the cross terms vanish by independence. Using

$$\text{Cov}(AG) = A\Sigma_G A^\top,$$

together with

$$\text{Cov}(\varepsilon) = \sigma^2 I_d,$$

gives

$$\text{Cov}(X, X) = A\Sigma_G A^\top + \sigma^2 I_d.$$

Therefore

$$\Sigma_X = A\Sigma_G A^\top + \sigma^2 I_d.$$

Collecting the covariance blocks yields

$$\text{Cov} \begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A\Sigma_G & A\Sigma_G A^\top + \sigma^2 I_d \end{pmatrix},$$

which proves the claim. $\square$

Equation (5) is the model's first interpretive statement: imaging covariance decomposes additively into a *genomics-driven* component $A\Sigma_G A^\top$ and an *irreducible* component $\sigma^2 I_d$. The cross-covariance $\Sigma_{GX} = \Sigma_G A^\top$ is the central estimable object — it names which genomic directions co-vary with imaging at all.

4 Bayes-optimal recovery of genomics from imaging

Proposition 2 (Posterior). *The posterior of genomics given imaging is Gaussian, $G \mid X \sim \mathcal{N}(\hat{g}(X), \Sigma_{G|X})$, with*

$$\hat{g}(X) = \mathbb{E}[G \mid X] = \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} X = BX, \quad (7)$$

$$\Sigma_{G|X} = \Sigma_G - \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} A \Sigma_G. \quad (8)$$

Equivalently, in precision form,

$$\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A, \quad B = \sigma^{-2} \Sigma_{G|X} A^\top. \quad (9)$$

Proof. Equations (7)–(8) are the standard Gaussian conditioning formulas applied to Proposition 1. The precision form follows from the Woodbury identity: $(\Sigma_G^{-1} + \sigma^{-2} A^\top A)^{-1} = \Sigma_G - \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} A \Sigma_G$, and then $B = \Sigma_G A^\top \Sigma_X^{-1} = \sigma^{-2} \Sigma_{G|X} A^\top$ by the push-through identity $\Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I)^{-1} = (\Sigma_G^{-1} + \sigma^{-2} A^\top A)^{-1} \sigma^{-2} A^\top$. $\square$

The estimator $\hat{g}(X) = BX$ is the Bayes-optimal predictor under squared-error loss; because the model is jointly Gaussian it is also the optimal predictor over *all* measurable functions of X , not merely linear ones. Thus $\Sigma_{G|X}$ is a hard floor on residual genomic uncertainty: **no algorithm can do better**. The precision form (9) makes the accounting transparent — imaging *adds* information $\sigma^{-2} A^\top A$ to the prior precision Σ_G^{-1} .

5 Mutual information and the SNR decomposition

Proposition 3 (Mutual information). *The mutual information between the two modalities is*

$$I(G; X) = \frac{1}{2} \log \frac{\det \Sigma_G}{\det \Sigma_{G|X}} = \frac{1}{2} \log \det (I_d + \sigma^{-2} A \Sigma_G A^\top). \quad (10)$$

Let $\lambda_1 \geq \dots \geq \lambda_r > 0$ be the nonzero eigenvalues of $A \Sigma_G A^\top$. Then

$$I(G; X) = \frac{1}{2} \sum_{i=1}^r \log \left(1 + \frac{\lambda_i}{\sigma^2} \right). \quad (11)$$

Proof. For jointly Gaussian (G, X) , $I(G; X) = \frac{1}{2} \log (\det \Sigma_G / \det \Sigma_{G|X})$. Substituting (9), $\det \Sigma_{G|X} = \det \Sigma_G / \det (I_p + \sigma^{-2} \Sigma_G^{1/2} A^\top A \Sigma_G^{1/2})$, and Sylvester’s determinant identity moves A to the other side to give $\det (I_d + \sigma^{-2} A \Sigma_G A^\top)$. Diagonalizing $A \Sigma_G A^\top$ yields (11). $\square$

Equation (11) is the cleanest reading of the model: each image-visible genomic direction contributes $\frac{1}{2} \log(1 + \text{SNR}_i)$ bits (nats), where $\text{SNR}_i = \lambda_i / \sigma^2$ is the signal-to-noise ratio of that direction. Directions whose genomic signal is small relative to imaging noise contribute essentially nothing, regardless of how many genes load on them.

6 Canonical coordinates and the recoverability spectrum

The eigenvalues λ_i are most interpretable after whitening. Define the whitened genomic variable $U = \Sigma_G^{-1/2}G \sim \mathcal{N}(0, I_p)$ and the *whitened transfer operator*

$$M := \sigma^{-1}A\Sigma_G^{1/2} \in \mathbb{R}^{d \times p}, \quad M = \sum_i d_i u_i v_i^\top \quad (\text{SVD}), \quad (12)$$

with left/right singular vectors $u_i \in \mathbb{R}^d$, $v_i \in \mathbb{R}^p$ and singular values $d_1 \geq d_2 \geq \dots \geq 0$.

Proposition 4 (Canonical decomposition of recoverability). *With M as in (12), define the genomic canonical direction $a_i := \Sigma_G^{-1/2}v_i$ and the canonical score $S_i := a_i^\top G = v_i^\top U$. Then:*

- (i) S_i has prior variance 1 and posterior variance $\text{Var}(S_i | X) = (1 + d_i^2)^{-1}$;
- (ii) the canonical correlation between S_i and imaging is $\rho_i = d_i / \sqrt{1 + d_i^2}$, equivalently $d_i^2 = \rho_i^2 / (1 - \rho_i^2)$;
- (iii) the eigenvalues of Section 5 satisfy $\lambda_i / \sigma^2 = d_i^2$, so

$$I(G; X) = \frac{1}{2} \sum_i \log(1 + d_i^2) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2). \quad (13)$$

Proof. In whitened coordinates the likelihood is $X | U \sim \mathcal{N}(\sigma MU, \sigma^2 I_d)$, since $AG = A\Sigma_G^{1/2}U = \sigma MU$. The posterior precision of U is $I_p + \sigma^{-2}(\sigma M)^\top(\sigma M) = I_p + M^\top M = V(I_p + D^2)V^\top$, so $\text{Var}(U | X) = V(I_p + D^2)^{-1}V^\top$ and $\text{Var}(S_i | X) = v_i^\top \text{Var}(U | X)v_i = (1 + d_i^2)^{-1}$, giving (i). For (ii), $\text{Cov}(U, X) = \mathbb{E}[U(\sigma MU + \varepsilon)^\top] = \sigma M^\top$, so $\text{Cov}(S_i, X) = \sigma v_i^\top M^\top = \sigma d_i u_i^\top$. The matched imaging variate $T_i := u_i^\top X$ has $\text{Var}(T_i) = u_i^\top \Sigma_X u_i = \sigma^2(1 + d_i^2)$ (using $\Sigma_X = \sigma^2(MM^\top + I_d)$) and $\text{Cov}(S_i, T_i) = \sigma d_i$, hence $\rho_i = \text{corr}(S_i, T_i) = d_i / \sqrt{1 + d_i^2}$. Part (iii) is immediate from $A\Sigma_G A^\top = \sigma^2 MM^\top$ and $\det(I_d + MM^\top) = \prod_i (1 + d_i^2)$. $\square$

Part (i) gives the headline identity. Define the recoverability of a genomic score as one minus the fraction of its variance that survives observing imaging (formalized in Section 7); then

$$R(a_i) = 1 - \text{Var}(S_i | X) = 1 - \frac{1}{1 + d_i^2} = \frac{d_i^2}{1 + d_i^2} = \rho_i^2. \quad (14)$$

The recoverability of the i -th canonical genomic direction is exactly the squared canonical correlation. This is what reduces the whole analysis to CCA between $\mathbf{G}$ and $\mathbf{X}$ (Section 10), and it means A and σ^2 never have to be estimated individually — only the three second moments $\Sigma_G, \Sigma_X, \Sigma_{GX}$ enter.

7 Recoverability of a named genomic direction

In practice one often cares about a specific, biologically named genomic score $Y = v^\top G$ for a fixed $v \in \mathbb{R}^p$ — a pathway signature, a polygenic score, a single gene's expression ($v = e_j$).

Definition 1 (Recoverability score). *For $v \in \mathbb{R}^p$ with $v^\top \Sigma_G v > 0$,*

$$R(v) := 1 - \frac{v^\top \Sigma_{G|X} v}{v^\top \Sigma_G v} = \frac{v^\top (\Sigma_G - \Sigma_{G|X}) v}{v^\top \Sigma_G v} \in [0, 1]. \quad (15)$$

$R(v) = 0$ means imaging carries no information about $Y = v^\top G$; $R(v) = 1$ means imaging determines it exactly. The score is scale-invariant in v and, by the Gaussian conditional-variance identity, equals the population R^2 of the Bayes-optimal predictor of Y from X : $R(v) = 1 - \mathbb{E}[\text{Var}(Y | X)] / \text{Var}(Y)$. The matrix at the core of (15) has the convenient moment-only form

$$\Sigma_G - \Sigma_{G|X} = \Sigma_G A^\top \Sigma_X^{-1} A \Sigma_G = \Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}. \quad (16)$$

8 The recoverability eigenproblem

The natural project-level question — *which* genomic directions does imaging see best — is the variational problem

$$\max_{v \neq 0} R(v) = \max_{v \neq 0} \frac{v^\top (\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}) v}{v^\top \Sigma_G v}. \quad (17)$$

Proposition 5 (Recoverability eigenproblem = CCA). *The stationary points of (17) are the solutions of the generalized eigenproblem*

$$\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG} v = \mu \Sigma_G v. \quad (18)$$

Its eigenvalues are $\mu_i = \rho_i^2 = d_i^2 / (1 + d_i^2)$ and its eigenvectors are the canonical genomic directions a_i of Proposition 4, Σ_G -orthonormal ($a_i^\top \Sigma_G a_j = \delta_{ij}$). Thus the ordered solutions of (18) are precisely the canonical correlation directions between G and X , and $\max_v R(v) = \rho_1^2$.

Proof. (18) is the first-order condition of the Rayleigh quotient (17). Substitute $v = \Sigma_G^{-1/2} w$ and $\Sigma_{GX} = \Sigma_G A^\top$; the generalized eigenproblem becomes the ordinary one $\Sigma_G^{1/2} A^\top \Sigma_X^{-1} A \Sigma_G^{1/2} w = \mu w$. Now $A \Sigma_G^{1/2} = \sigma M$ and $\Sigma_X = \sigma^2 (M M^\top + I_d)$, so

$$\Sigma_G^{1/2} A^\top \Sigma_X^{-1} A \Sigma_G^{1/2} = \sigma M^\top \cdot \sigma^{-2} (M M^\top + I_d)^{-1} \cdot \sigma M = M^\top (M M^\top + I_d)^{-1} M.$$

With the SVD $M = U D V^\top$ this equals $V D^2 (D^2 + I)^{-1} V^\top$, whose eigenvalues are $d_i^2 / (1 + d_i^2) = \rho_i^2$ with eigenvectors v_i . Hence $w = v_i$ and $v = \Sigma_G^{-1/2} v_i = a_i$; the Σ_G -orthonormality $a_i^\top \Sigma_G a_j = v_i^\top v_j = \delta_{ij}$ is immediate. $\square$

So the deliverable of the analysis is a *recoverability spectrum* $\rho_1^2 \geq \rho_2^2 \geq \dots$ together with the genomic directions a_i that achieve it. Reading the loadings of $a_1, a_2, \dots$ back onto genes or pathways answers “which biology is image-identifiable”: the leading directions typically align with coarse, spatially expressed programs — proliferation, hypoxia, angiogenesis, necrosis, immune infiltration, tumor burden, stromal remodeling — rather than with individual transcripts.

9 Rank limits and identifiability

Proposition 6 (Rank limit). *The recoverable signal matrix has rank*

$$\text{rank}(\Sigma_G - \Sigma_{G|X}) = \text{rank}(A \Sigma_G A^\top) \leq \min(d, p, \text{rank } A, \text{rank } \Sigma_G). \quad (19)$$

Consequently at most $\min(d, p, \text{rank } A, \text{rank } \Sigma_G)$ canonical correlations are nonzero, and the image-identifiable genomic subspace $\text{span}\{a_i : \rho_i > 0\}$ has dimension bounded by the same quantity.

This is the conceptual core. Even with $p = 20,000$ measured genes, imaging can expose only a projection of genomics whose dimension is capped by the rank of the transfer map A — if a CT phenotype reflects, say, 20 macroscopic biological axes, then no model recovers 20,000 independent molecular variables from it. The recoverable dimension is a property of the *biology–imaging channel*, not of the estimator. A data-processing remark: imaging features are themselves a deterministic, lossy function $x = \phi(\text{image})$; any such post-processing can only shrink $\text{rank}(\Sigma_{GX})$ and lower each ρ_i , never raise them.

10 Estimation from finite samples

In applications p and d may rival or exceed n , the noise is not isotropic, and A, Σ_G, σ^2 are unknown. Because Propositions 4 and 5 require only $\Sigma_G, \Sigma_X, \Sigma_{GX}$, we estimate those directly.

10.1 Regularized covariances

Replace sample covariances by shrinkage estimators (e.g. Ledoit–Wolf), $\hat{\Sigma}_G = (1 - \alpha_G) S_G + \alpha_G \nu_G I_p$ and likewise $\hat{\Sigma}_X$, with $S_G = n^{-1} \mathbf{G}^\top \mathbf{G}$ the centered sample covariance; the cross-covariance is $\hat{\Sigma}_{GX} = n^{-1} \mathbf{G}^\top \mathbf{X}$. Shrinkage keeps $\hat{\Sigma}_G, \hat{\Sigma}_X$ invertible and well-conditioned, which is required for the whitening in (12). When p is very large it is practical to first project genomics onto its top principal components (or onto curated pathway scores), reducing p to a manageable $p' \ll n$ before estimation; the rank limit of Proposition 6 means little is lost provided p' exceeds the recoverable dimension.

10.2 Reduced-rank and ridge regression

The transfer map, if needed explicitly, is the multivariate regression of X on G . Ridge gives $\hat{A}^\top = (\mathbf{G}^\top \mathbf{G} + \gamma I_p)^{-1} \mathbf{G}^\top \mathbf{X}$, and $\hat{\sigma}^2$ follows from the residual covariance. Imposing $\text{rank}(A) \leq k$ (reduced-rank regression) bakes Proposition 6 into the fit and is equivalent, up to regularization, to truncating the SVD of M at k components.

10.3 Regularized CCA

The estimator used in the companion notebook solves (18) with the regularized covariances of Section 10.1: form $\hat{M} = \hat{\Sigma}_G^{-1/2} \hat{\Sigma}_{GX} \hat{\Sigma}_X^{-1/2}$, take its SVD $\hat{M} = \hat{U} \hat{D} \hat{V}^\top$, and read off $\hat{\rho}_i = \hat{d}_i$ (clipped to $[0, 1]$), $\hat{R}_i = \hat{\rho}_i^2$, genomic directions $\hat{a}_i = \hat{\Sigma}_G^{-1/2} \hat{u}_i$, imaging directions $\hat{b}_i = \hat{\Sigma}_X^{-1/2} \hat{v}_i$. The Bayes-optimal predictor and posterior covariance follow from (7)–(8) with hats throughout. This is exactly regularized CCA — the recoverability program and CCA are the same computation.

10.4 Probabilistic CCA: the latent-variable view

The rank limit motivates an explicit low-dimensional generative form. With a shared latent $Z \in \mathbb{R}^k$,

$$Z \sim \mathcal{N}(0, I_k), \quad G = W_G Z + \mu_G + \epsilon_G, \quad X = W_X Z + \mu_X + \epsilon_X, \quad (20)$$

$\epsilon_G \sim \mathcal{N}(0, \Psi_G)$, $\epsilon_X \sim \mathcal{N}(0, \Psi_X)$ independent (probabilistic CCA; Bach & Jordan, 2005). Here k is the dimension of the image-identifiable genomic subspace, Z is the shared biological state, and ϵ_G, ϵ_X are modality-private variation. The implied moments are $\Sigma_G = W_G W_G^\top + \Psi_G$, $\Sigma_X = W_X W_X^\top + \Psi_X$, $\Sigma_{GX} = W_G W_X^\top$; the maximum-likelihood loadings are the canonical directions scaled by the canonical correlations, recovering Section 10.3. The model is fit by EM, which also yields posterior latent estimates $\mathbb{E}[Z | x]$ and naturally accommodates anisotropic (per-feature) noise, unlike the isotropic $\sigma^2 I_d$ of the base model.

10.5 Sample-size regimes and dimensional collapse

Equations (7)–(18) are population identities; the estimators of Sections 10.1–10.4 inherit a finite-sample bias whose magnitude is governed entirely by the aspect ratios p/n and d/n . The bias is not benign, and below a critical sample size the recoverability spectrum breaks down in a specific, predictable way.

Three regimes. Let p^*, d^* be the working dimensions after any pre-reduction (e.g. PCA), and write $\kappa_G = p^*/n$, $\kappa_X = d^*/n$.

1. *Classical* ($\kappa_G + \kappa_X \ll 1$). Sample CCA is consistent; $\hat{\rho}_i \rightarrow \rho_i$ at the usual $n^{-1/2}$ rate and the in-sample spectrum is a faithful estimate of the population one.
2. *High-dimensional* ($\kappa_G + \kappa_X \lesssim 1$, both bounded away from 1). Under the null $\Sigma_{GX} = 0$, the squared sample canonical correlations follow a Wachter-type law on $[(\sqrt{\kappa_G(1-\kappa_X)} - \sqrt{\kappa_X(1-\kappa_G)})^2, (\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2]$ (Wachter, 1980; Yang & Pan, 2015); the leading sample value already sits well above zero by geometry. Estimates of ρ_i are systematically inflated and ordinary CCA must be replaced by its regularized form, with shrinkage parameters tuned by cross-validation.
3. *Saturating* ($p^* + d^* \geq n$). The rows of $\mathbf{G}$ and $\mathbf{X}$ live in subspaces whose intersection has dimension at least $p^* + d^* - n + 1$; consequently $\hat{\rho}_1 = \dots = \hat{\rho}_q = 1$ for $q = \min(p^*, d^*) - (n - \max(p^*, d^*))_+$ regardless of any underlying association. The empirical mutual information diverges, the in-sample $\text{tr}(\hat{\Sigma}_{GX}\hat{\Sigma}_X^{-1}\hat{\Sigma}_{XG})/\text{tr}(\hat{\Sigma}_X)$ approaches the rank fraction of the genomic span, and the permutation null collapses onto the observed value (its p -values concentrate near 1). Nothing in the spectrum is estimable in this regime without shrinkage *plus* dimension reduction; shrinkage alone keeps the matrices invertible but does not restore identifiability.

Quantitative warning signs. Three diagnostics, jointly, separate real signal from dimensional artefact:

- the gap between in-sample $\hat{R}_i$ and the K -fold value $\hat{R}_i^{\text{cv}}$; under regime 3 this gap approaches $\hat{R}_i - 0$;
- the permutation null for $\hat{\rho}_1$: if the observed value lies inside the bulk, the result is consistent with chance;
- the cross-validated R^2 of the leading canonical score $a_1^\top g$ predicted from x on a held-out split; in regime 3 it is typically large and *negative* (worse than predicting the mean), a behaviour seen empirically on TCGA-KIRC at $n = 131$ with $p^* = 100, d^* = 107$ in the companion notebook.

Recommendations. Choose p^* and d^* *adaptively from n* , not from the native modality dimensions:

- Target $p^* + d^* \leq n/\tau$ with $\tau \geq 5$ as a working rule (we use $p^*, d^* \leq \lfloor n/\tau \rfloor$ per side in the notebook; $\tau \in [5, 10]$ trades bias for resolution). Reduce both modalities by PCA when needed — reducing only the larger side leaves the estimator in regime 3.
- Treat $\hat{R}_i^{\text{cv}}$ (Section 10.6) as the primary estimand and report the in-sample $\hat{R}_i$ only with the gap made explicit; the in-sample fraction of imaging variance attributable to genomics should be reported alongside a cross-validated counterpart, e.g. $\sum_i \hat{R}_i^{\text{cv}} / \min(p^*, d^*)$.

- Power planning. The leading population correlation ρ_1 is detectable above the Wachter bulk only when its squared value exceeds $(\sqrt{\kappa_G(1 - \kappa_X)} + \sqrt{\kappa_X(1 - \kappa_G)})^2$; solving for n gives a minimum sample size given a hypothesised effect. For $\rho_1^2 = 0.2$ and a balanced $p^* = d^*$, this requires roughly $n \gtrsim 25 p^*$, an order of magnitude beyond what cohort sizes in radiogenomics usually offer — reinforcing the need for aggressive pre-reduction.
- Probabilistic CCA (Section 10.4) with a small, fixed latent dimension k is the most parsimonious option when n is the binding constraint; rank- k regularization sidesteps the high-dimensional bulk by construction.

The practical message is conservative: in radiogenomic cohorts, sample size is almost always the binding constraint and a defensible analysis is the one whose effective $p^* + d^*$ is a small fraction of n , with both sides reduced symmetrically and every reported number a cross-validated or permutation-calibrated one.

10.6 Honest recoverability: cross-validation and permutation

In-sample canonical correlations are biased upward, severely so when p, d are not small relative to n . Two safeguards, both implemented in the notebook:

- *Cross-validation.* Estimate $\hat{a}_i, \hat{b}_i$ on a training fold; on the held-out fold form the scores $\hat{a}_i^\top g$ and $\hat{b}_i^\top x$ and report their out-of-sample squared correlation as the honest $\hat{R}_i$. The gap to the in-sample value quantifies overfitting.
- *Permutation test.* Repeatedly permute the patient labels of one modality, refit, and collect the leading canonical correlation to build a null distribution; the permutation p -value tests whether any genomic direction is image-identifiable beyond chance.

11 Interpretation for radiogenomics

The model reframes the radiogenomic question. Rather than testing whether CT predicts each gene — a high-dimensional, low-power multiple-testing problem — it asks for the *recoverability spectrum* $\{\rho_i^2\}$ and the associated genomic directions $\{a_i\}$. The outputs are: (1) a small number of image-identifiable genomic axes, with gene/pathway loadings that make them biologically legible; (2) a hard bound $\Sigma_{G|X}$ on what any predictor can achieve, per direction; (3) a total information budget $I(G; X)$; and (4) an honest, cross-validated estimate of each. The governing principle is that radiogenomics should target *image-identifiable low-dimensional genomic subspaces*, whose dimension is set by the biology–imaging channel and not by the gene panel size.

A Symbol reference

$g \in \mathbb{R}^p, x \in \mathbb{R}^d$	per-patient genomic / imaging feature vectors
$\mathbf{G} \in \mathbb{R}^{n \times p}, \mathbf{X} \in \mathbb{R}^{n \times d}$	stacked data matrices (n patients)
$A \in \mathbb{R}^{d \times p}$	genomics-to-imaging transfer map
σ^2	isotropic imaging noise variance
$\Sigma_G, \Sigma_X, \Sigma_{GX}$	genomic / imaging / cross covariance
$\Sigma_{G X}$	posterior covariance of G given X (recoverability floor)
$B = \Sigma_{GX} \Sigma_X^{-1}$	Bayes-optimal linear map $x \mapsto \mathbb{E}[g \mid x]$
$M = \sigma^{-1} A \Sigma_G^{-1/2}$	whitened transfer operator; d_i its singular values
$\rho_i, R_i = \rho_i^2$	i -th canonical correlation / recoverability
a_i, b_i	i -th canonical genomic / imaging direction
$R(v)$	recoverability score of a named genomic direction v
$I(G; X)$	mutual information between modalities
k	dimension of the image-identifiable genomic subspace